เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1

Subject: 394173 Mathematics III

Date: 11 December 2014

Year: 2014

Section: 15-16

Time: 13.00-16.00

Name:___________________________ ID:_________________________ Class:____________

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. **No** documents are allowed to be taken out of the examination room.
3. Textbooks, dictionary and calculators are **NOT** allowed.
4. This is a closed book examination.
5. Electronic communication device are **NOT** allowed in the examination room.
6. The examination has 9 pages (including this page), 9 questions and 1 part. The total score is 39 points.
7. Write all your answers on this examination sheet.

No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	Total score

1. The first term of an arithmetic sequence is 2 and let a_4, a_{11}, a_{18} be the first 3 terms of geometric sequence. Find the sum of the first 3 terms of this geometric sequence.

(4 points)

2. Find each limit, or state that the limit does not exist and explain your reasoning.

2.1 $a_n = \sqrt{n^2+5} - n$ (2 points)

2.2 $a_n = (-1)^{n+3}\left(\dfrac{2n^2-5n+1}{\sqrt{1+4n^4}} + \dfrac{3}{\sqrt{n^2+2}}\right)$ (2 points)

3. The 7^{th} term of an arithmetic sequence is 25, and the 12^{th} term is 45.

Find $\sum_{i=1}^{10}(a_i + i^2)$? (4 points)

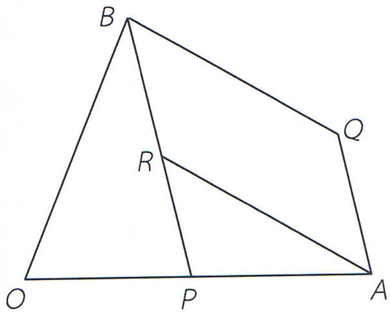
4. In the geometric series, if $a_6 = 80$ and $\sum_{n=4}^{6} a_n = 65$. Find a_1 and r. (4 points)

5. Given $S = 1 + \frac{2}{5} + \frac{3}{25} + \frac{4}{125} + \ldots$ and $16S - n = \frac{1}{1\cdot3} + \frac{1}{3\cdot5} + \frac{1}{5\cdot7} + \ldots.$

Find the value of n. (5 points)

6. From the figure, let $\overrightarrow{OA} = \vec{a}$ and $\overrightarrow{OB} = \vec{b}$ such that $\overrightarrow{OP} : \overrightarrow{PA} = 1 : 2$ and $\overrightarrow{PR} : \overrightarrow{RB} = 1 : 2$. If ARBQ is the parallelogram, find $\overrightarrow{OQ}$ in the form $\vec{a}$ and $\vec{b}$?

(5 points)



7. Assume that $\vec{u} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{v} = -2\hat{i} + \hat{j} + 3\hat{k}$. Find the vector has length as $\vec{u}\cdot\vec{v}$ which has the same direction as $\vec{u} - 2\vec{v}$? (4 points)

8. Assume that $\vec{u} = 2\hat{i} - 5\hat{j}$, $\vec{v} = \hat{i} + 2\hat{j}$ and $\vec{w}$ is a vector such that $\vec{u}\cdot\vec{w} = -11$ and $\vec{v}\cdot\vec{w} = 8$. If θ is the angle between $\vec{w}$ and $5\hat{i} + \hat{j}$, find $\tan\theta + \sin 2\theta$?

(5 points)

9. Let $A(2, 0, 3)$, $B(2, 4, 5)$ and $C(1, x, 3)$ be the points, where x is a positive integer such that the area of triangle $\triangle ABC = 3$ units^2. Find $\overrightarrow{BC} \cdot \overrightarrow{AC}$ (4 points)